

(SR\_venv) PS  
C:\Universidad\4\_ano\segundo\_cuatri\sistemas\_recomendacion\practica\_sistemas\_recomendacion> &  
c:\Universidad\4\_ano\segundo\_cuatri\sistemas\_recomendacion\practica\_sistemas\_recomendacion\SR\_venv  
\Scripts\python.exe  
c:/Universidad/4\_ano/segundo\_cuatri/sistemas\_recomendacion/practica\_sistemas\_recomendacion/code/n  
eighborhood-based\_recommendation.py  
Shape matriz: (990000, 2262292)  
NNZ: 64767209  
Construcción completada en 169.06s  
Playlists test: 10000

=====

=====

==

## EJECUTANDO MODELO PLAYLIST-PLAYLIST

---

[playlist-playlist] chunks: 20 | chunk\_size: 500 | workers: 4  
[playlist-playlist] 1/20 | 500/10000 playlists | 10.10s transcurridos | 3.20 min restantes  
[playlist-playlist] 2/20 | 1000/10000 playlists | 10.10s transcurridos | 1.52 min restantes  
[playlist-playlist] 3/20 | 1500/10000 playlists | 23.58s transcurridos | 2.23 min restantes  
[playlist-playlist] 4/20 | 2000/10000 playlists | 23.70s transcurridos | 1.58 min restantes  
[playlist-playlist] 5/20 | 2500/10000 playlists | 26.68s transcurridos | 1.33 min restantes  
[playlist-playlist] 6/20 | 3000/10000 playlists | 29.37s transcurridos | 1.14 min restantes  
[playlist-playlist] 7/20 | 3500/10000 playlists | 46.99s transcurridos | 1.45 min restantes  
[playlist-playlist] 8/20 | 4000/10000 playlists | 47.31s transcurridos | 1.18 min restantes  
[playlist-playlist] 9/20 | 4500/10000 playlists | 50.52s transcurridos | 1.03 min restantes  
[playlist-playlist] 10/20 | 5000/10000 playlists | 53.51s transcurridos | 0.89 min restantes  
[playlist-playlist] 11/20 | 5500/10000 playlists | 84.19s transcurridos | 1.15 min restantes  
[playlist-playlist] 12/20 | 6000/10000 playlists | 84.38s transcurridos | 0.94 min restantes  
[playlist-playlist] 13/20 | 6500/10000 playlists | 87.24s transcurridos | 0.78 min restantes  
[playlist-playlist] 14/20 | 7000/10000 playlists | 89.08s transcurridos | 0.64 min restantes  
[playlist-playlist] 15/20 | 7500/10000 playlists | 181.62s transcurridos | 1.01 min restantes  
[playlist-playlist] 16/20 | 8000/10000 playlists | 182.73s transcurridos | 0.76 min restantes  
[playlist-playlist] 17/20 | 8500/10000 playlists | 184.46s transcurridos | 0.54 min restantes  
[playlist-playlist] 18/20 | 9000/10000 playlists | 185.79s transcurridos | 0.34 min restantes  
[playlist-playlist] 19/20 | 9500/10000 playlists | 193.96s transcurridos | 0.17 min restantes  
[playlist-playlist] 20/20 | 10000/10000 playlists | 194.68s transcurridos | 0.00 min restantes

=====

=====

==

# EJECUTANDO MODELO TRACK-TRACK PRECOMPUTED

---

Tracks seed únicos: 62810

[track-track precompute] chunks: 63 | chunk\_size: 1000 | workers: 6

[track-track precompute] 1/63 | 1000/62810 seeds | 89.47s transcurridos | 92.17 min restantes

[track-track precompute] 2/63 | 2000/62810 seeds | 90.38s transcurridos | 45.80 min restantes

[track-track precompute] 3/63 | 3000/62810 seeds | 93.39s transcurridos | 31.03 min restantes

[track-track precompute] 4/63 | 4000/62810 seeds | 95.94s transcurridos | 23.51 min restantes

[track-track precompute] 5/63 | 5000/62810 seeds | 102.36s transcurridos | 19.72 min restantes

[track-track precompute] 6/63 | 6000/62810 seeds | 102.80s transcurridos | 16.22 min restantes

[track-track precompute] 7/63 | 7000/62810 seeds | 164.26s transcurridos | 21.83 min restantes

[track-track precompute] 8/63 | 8000/62810 seeds | 169.16s transcurridos | 19.32 min restantes

[track-track precompute] 9/63 | 9000/62810 seeds | 169.25s transcurridos | 16.87 min restantes

[track-track precompute] 10/63 | 10000/62810 seeds | 171.06s transcurridos | 15.06 min restantes

[track-track precompute] 11/63 | 11000/62810 seeds | 171.52s transcurridos | 13.46 min restantes

[track-track precompute] 12/63 | 12000/62810 seeds | 172.22s transcurridos | 12.15 min restantes

[track-track precompute] 13/63 | 13000/62810 seeds | 221.85s transcurridos | 14.17 min restantes

[track-track precompute] 14/63 | 14000/62810 seeds | 222.51s transcurridos | 12.93 min restantes

[track-track precompute] 15/63 | 15000/62810 seeds | 224.23s transcurridos | 11.91 min restantes

[track-track precompute] 16/63 | 16000/62810 seeds | 224.43s transcurridos | 10.94 min restantes

[track-track precompute] 17/63 | 17000/62810 seeds | 227.70s transcurridos | 10.23 min restantes

[track-track precompute] 18/63 | 18000/62810 seeds | 228.59s transcurridos | 9.48 min restantes

[track-track precompute] 19/63 | 19000/62810 seeds | 263.16s transcurridos | 10.11 min restantes

[track-track precompute] 20/63 | 20000/62810 seeds | 263.50s transcurridos | 9.40 min restantes

[track-track precompute] 21/63 | 21000/62810 seeds | 264.95s transcurridos | 8.79 min restantes

[track-track precompute] 22/63 | 22000/62810 seeds | 266.36s transcurridos | 8.23 min restantes

[track-track precompute] 23/63 | 23000/62810 seeds | 266.47s transcurridos | 7.69 min restantes

[track-track precompute] 24/63 | 24000/62810 seeds | 267.83s transcurridos | 7.22 min restantes

[track-track precompute] 25/63 | 25000/62810 seeds | 293.22s transcurridos | 7.39 min restantes

[track-track precompute] 26/63 | 26000/62810 seeds | 294.33s transcurridos | 6.95 min restantes

[track-track precompute] 27/63 | 27000/62810 seeds | 295.82s transcurridos | 6.54 min restantes

[track-track precompute] 28/63 | 28000/62810 seeds | 295.91s transcurridos | 6.13 min restantes

[track-track precompute] 29/63 | 29000/62810 seeds | 296.19s transcurridos | 5.76 min restantes

[track-track precompute] 30/63 | 30000/62810 seeds | 296.49s transcurridos | 5.40 min restantes

[track-track precompute] 31/63 | 31000/62810 seeds | 315.04s transcurridos | 5.39 min restantes

[track-track precompute] 32/63 | 32000/62810 seeds | 317.23s transcurridos | 5.09 min restantes

[track-track precompute] 33/63 | 33000/62810 seeds | 317.67s transcurridos | 4.78 min restantes

[track-track precompute] 34/63 | 34000/62810 seeds | 317.72s transcurridos | 4.49 min restantes

[track-track precompute] 35/63 | 35000/62810 seeds | 318.17s transcurridos | 4.21 min restantes

[track-track precompute] 36/63 | 36000/62810 seeds | 320.40s transcurridos | 3.98 min restantes

[track-track precompute] 37/63 | 37000/62810 seeds | 332.49s transcurridos | 3.87 min restantes

[track-track precompute] 38/63 | 38000/62810 seeds | 333.39s transcurridos | 3.63 min restantes

[track-track precompute] 39/63 | 39000/62810 seeds | 334.30s transcurridos | 3.40 min restantes

[track-track precompute] 40/63 | 40000/62810 seeds | 334.63s transcurridos | 3.18 min restantes  
 [track-track precompute] 41/63 | 41000/62810 seeds | 334.87s transcurridos | 2.97 min restantes  
 [track-track precompute] 42/63 | 42000/62810 seeds | 335.28s transcurridos | 2.77 min restantes  
 [track-track precompute] 43/63 | 43000/62810 seeds | 344.26s transcurridos | 2.64 min restantes  
 [track-track precompute] 44/63 | 44000/62810 seeds | 344.73s transcurridos | 2.46 min restantes  
 [track-track precompute] 45/63 | 45000/62810 seeds | 345.21s transcurridos | 2.28 min restantes  
 [track-track precompute] 46/63 | 46000/62810 seeds | 345.49s transcurridos | 2.10 min restantes  
 [track-track precompute] 47/63 | 47000/62810 seeds | 345.53s transcurridos | 1.94 min restantes  
 [track-track precompute] 48/63 | 48000/62810 seeds | 345.57s transcurridos | 1.78 min restantes  
 [track-track precompute] 49/63 | 49000/62810 seeds | 351.10s transcurridos | 1.65 min restantes  
 [track-track precompute] 50/63 | 50000/62810 seeds | 351.27s transcurridos | 1.50 min restantes  
 [track-track precompute] 51/63 | 51000/62810 seeds | 352.03s transcurridos | 1.36 min restantes  
 [track-track precompute] 52/63 | 52000/62810 seeds | 352.49s transcurridos | 1.22 min restantes  
 [track-track precompute] 53/63 | 53000/62810 seeds | 352.53s transcurridos | 1.09 min restantes  
 [track-track precompute] 54/63 | 54000/62810 seeds | 352.72s transcurridos | 0.96 min restantes  
 [track-track precompute] 55/63 | 55000/62810 seeds | 354.71s transcurridos | 0.84 min restantes  
 [track-track precompute] 56/63 | 56000/62810 seeds | 355.16s transcurridos | 0.72 min restantes  
 [track-track precompute] 57/63 | 57000/62810 seeds | 355.44s transcurridos | 0.60 min restantes  
 [track-track precompute] 58/63 | 58000/62810 seeds | 355.54s transcurridos | 0.49 min restantes  
 [track-track precompute] 59/63 | 59000/62810 seeds | 355.63s transcurridos | 0.38 min restantes  
 [track-track precompute] 60/63 | 60000/62810 seeds | 355.84s transcurridos | 0.28 min restantes  
 [track-track precompute] 61/63 | 61000/62810 seeds | 356.08s transcurridos | 0.18 min restantes  
 [track-track precompute] 62/63 | 62000/62810 seeds | 356.12s transcurridos | 0.08 min restantes  
 [track-track precompute] 63/63 | 62810/62810 seeds | 356.16s transcurridos | 0.00 min restantes  
 [track-track recommend] chunks: 40 | chunk\_size: 250 | workers: 4  
 [track-track recommend] 1/40 | 250/10000 playlists | 47.75s transcurridos | 31.04 min restantes  
 [track-track recommend] 2/40 | 500/10000 playlists | 47.75s transcurridos | 15.12 min restantes  
 [track-track recommend] 3/40 | 750/10000 playlists | 47.75s transcurridos | 9.82 min restantes  
 [track-track recommend] 4/40 | 1000/10000 playlists | 47.75s transcurridos | 7.16 min restantes  
 [track-track recommend] 5/40 | 1250/10000 playlists | 47.75s transcurridos | 5.57 min restantes  
 [track-track recommend] 6/40 | 1500/10000 playlists | 47.76s transcurridos | 4.51 min restantes  
 [track-track recommend] 7/40 | 1750/10000 playlists | 47.76s transcurridos | 3.75 min restantes  
 [track-track recommend] 8/40 | 2000/10000 playlists | 47.76s transcurridos | 3.18 min restantes  
 [track-track recommend] 9/40 | 2250/10000 playlists | 47.76s transcurridos | 2.74 min restantes  
 [track-track recommend] 10/40 | 2500/10000 playlists | 47.86s transcurridos | 2.39 min restantes  
 [track-track recommend] 11/40 | 2750/10000 playlists | 47.87s transcurridos | 2.10 min restantes  
 [track-track recommend] 12/40 | 3000/10000 playlists | 47.88s transcurridos | 1.86 min restantes  
 [track-track recommend] 13/40 | 3250/10000 playlists | 48.21s transcurridos | 1.67 min restantes  
 [track-track recommend] 14/40 | 3500/10000 playlists | 48.22s transcurridos | 1.49 min restantes  
 [track-track recommend] 15/40 | 3750/10000 playlists | 48.23s transcurridos | 1.34 min restantes  
 [track-track recommend] 16/40 | 4000/10000 playlists | 48.38s transcurridos | 1.21 min restantes  
 [track-track recommend] 17/40 | 4250/10000 playlists | 48.40s transcurridos | 1.09 min restantes  
 [track-track recommend] 18/40 | 4500/10000 playlists | 48.41s transcurridos | 0.99 min restantes  
 [track-track recommend] 19/40 | 4750/10000 playlists | 48.53s transcurridos | 0.89 min restantes  
 [track-track recommend] 20/40 | 5000/10000 playlists | 48.56s transcurridos | 0.81 min restantes

[track-track recommend] 21/40 | 5250/10000 playlists | 48.66s transcurridos | 0.73 min restantes  
[track-track recommend] 22/40 | 5500/10000 playlists | 48.68s transcurridos | 0.66 min restantes  
[track-track recommend] 23/40 | 5750/10000 playlists | 48.82s transcurridos | 0.60 min restantes  
[track-track recommend] 24/40 | 6000/10000 playlists | 48.96s transcurridos | 0.54 min restantes  
[track-track recommend] 25/40 | 6250/10000 playlists | 48.97s transcurridos | 0.49 min restantes  
[track-track recommend] 26/40 | 6500/10000 playlists | 49.05s transcurridos | 0.44 min restantes  
[track-track recommend] 27/40 | 6750/10000 playlists | 49.16s transcurridos | 0.39 min restantes  
[track-track recommend] 28/40 | 7000/10000 playlists | 49.26s transcurridos | 0.35 min restantes  
[track-track recommend] 29/40 | 7250/10000 playlists | 49.67s transcurridos | 0.31 min restantes  
[track-track recommend] 30/40 | 7500/10000 playlists | 49.67s transcurridos | 0.28 min restantes  
[track-track recommend] 31/40 | 7750/10000 playlists | 49.76s transcurridos | 0.24 min restantes  
[track-track recommend] 32/40 | 8000/10000 playlists | 49.89s transcurridos | 0.21 min restantes  
[track-track recommend] 33/40 | 8250/10000 playlists | 50.26s transcurridos | 0.18 min restantes  
[track-track recommend] 34/40 | 8500/10000 playlists | 50.31s transcurridos | 0.15 min restantes  
[track-track recommend] 35/40 | 8750/10000 playlists | 50.34s transcurridos | 0.12 min restantes  
[track-track recommend] 36/40 | 9000/10000 playlists | 50.35s transcurridos | 0.09 min restantes  
[track-track recommend] 37/40 | 9250/10000 playlists | 50.38s transcurridos | 0.07 min restantes  
[track-track recommend] 38/40 | 9500/10000 playlists | 50.39s transcurridos | 0.04 min restantes  
[track-track recommend] 39/40 | 9750/10000 playlists | 50.44s transcurridos | 0.02 min restantes  
[track-track recommend] 40/40 | 10000/10000 playlists | 50.54s transcurridos | 0.00 min restantes

=====

=====

==

## COMPARATIVA FINAL

---

### PLAYLIST-PLAYLIST

Tiempo modelo : 194.94s

R-Precision : 0.157477

NDCG@500 : 0.346263

Clicks : 4.467800

### TRACK-TRACK PRECOMPUTED

Tiempo precompute : 356.28s

Tiempo recommend : 51.56s

Tiempo total modelo : 407.84s

R-Precision : 0.131366

NDCG@500 : 0.289358

Clicks : 6.157600

### GANADOR

Por calidad general: PLAYLIST-PLAYLIST

## DIFERENCIAS

$\Delta$  R-Precision : 0.026111

$\Delta$  NDCG@500 : 0.056905

$\Delta$  Clicks : -1.689800

$\Delta$  Tiempo total : -212.90s